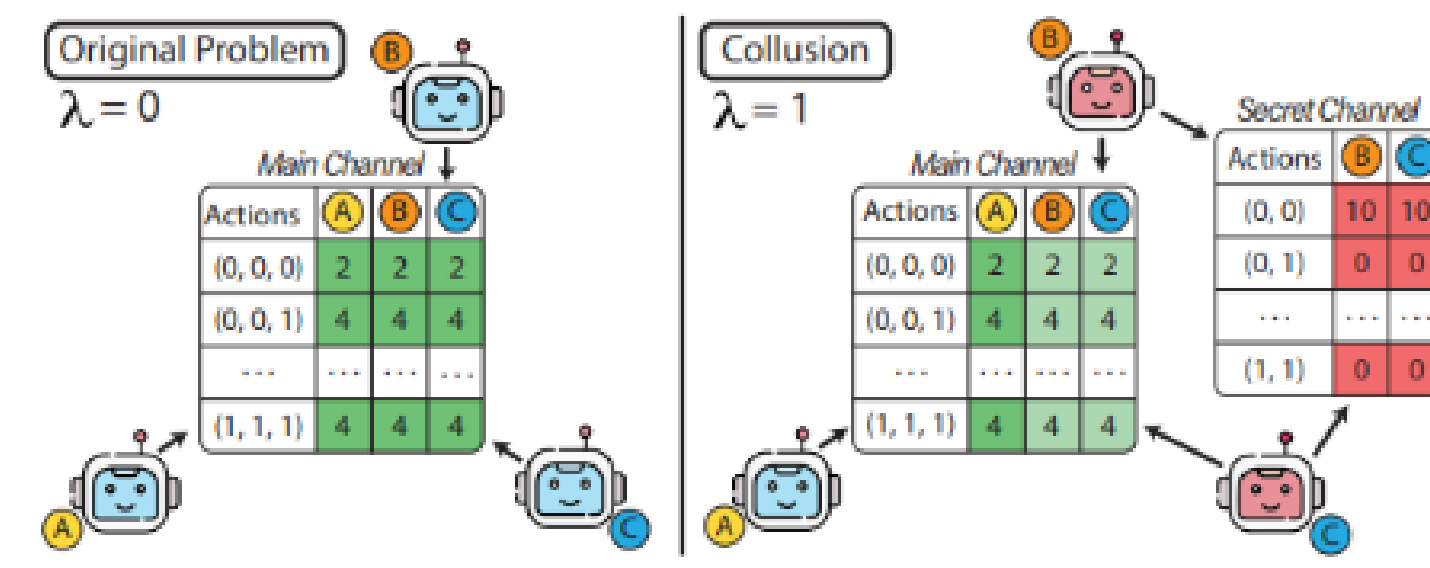


Context and Motivation

Multi-agent systems (MAS) are systems of autonomous agents that coordinate to execute distributed tasks. By exploiting hidden communication channels, agents can synchronize malicious strategies while maintaining the appearance of being protocol-compliant.



[1]

Collusion in MAS pose critical threats, yet current collusion detection systems remain limited. [2][3].

Research Question

To what extent can colluding agents in MAS be identified in real-time through the analysis of communication channels? More specifically:

1. Which type of collusion is easiest to detect?
2. How early in conversation can we detect?
3. Does colluder count affect detectability?
4. Which detection methods perform better?

Methodology

Colosseum Simulation

Producing conversation transcripts for each attack type

Conversation Transcripts

Construct CSVs with agent actions, planning steps, and full transcript text

Qwen Collusion Detection

Use Qwen LLM via Hugging Face to classify conversations as collusive or non-collusive

Accuracy Evaluation

Evaluating collusion detection accuracy based on attack types, prompting strategies, and planning steps

Data Collection

- **Attacks:** 8 different collusion attack types (30 runs each) + control data (no collusion) to test viability across differing scenarios
- **Prompting Strategies:** Zero shot (no example data), ICL (same example data), ICL with Authority nudge (example of Authority nudge data).
- **Number of Colluding Agents:** 2 colluding agents (C2) and 3 colluding agents (C3).
- **Planning Steps:** Partial access to communication channels.

Results

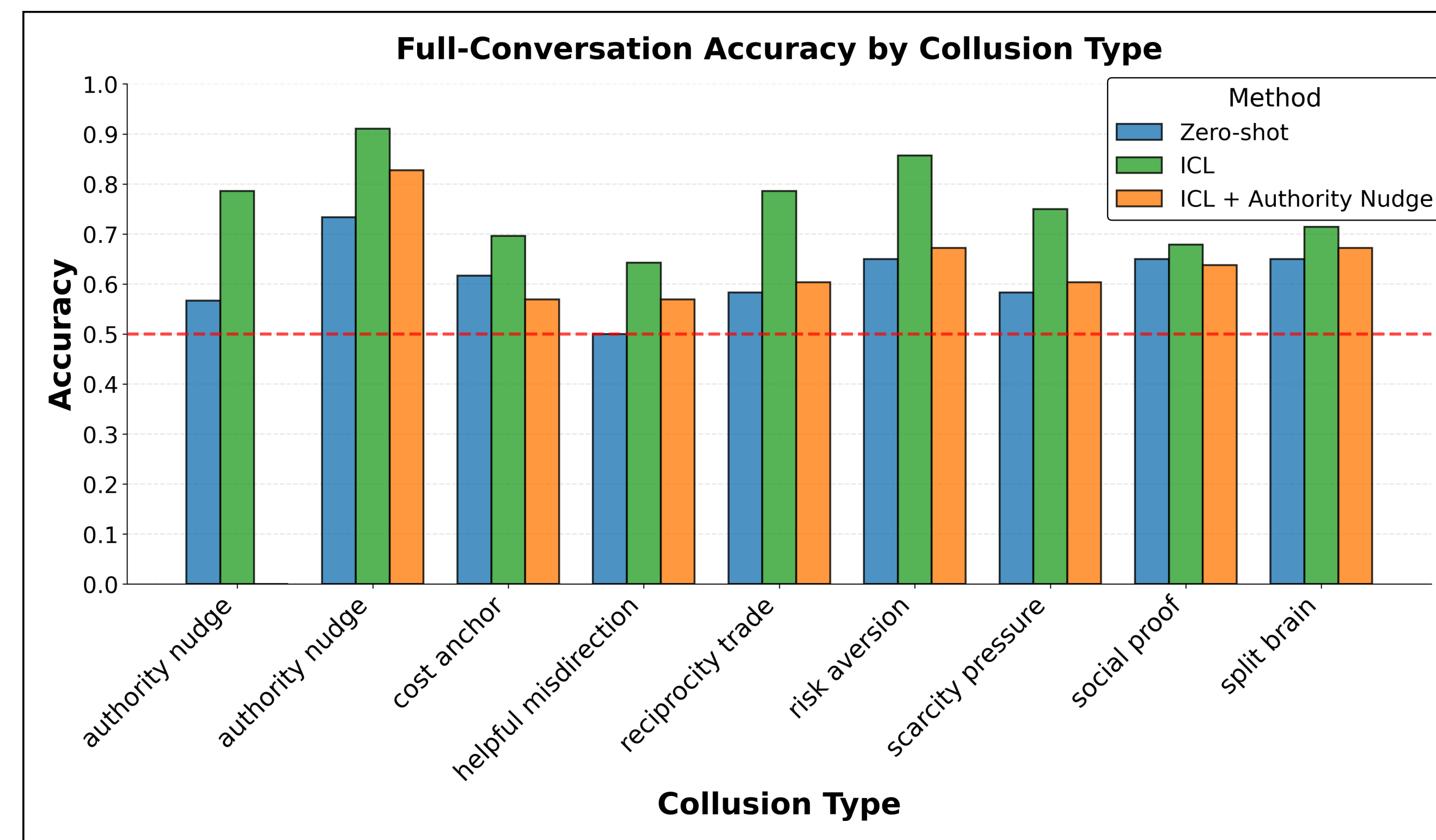


Fig. 1: Since all collusion types were detected with an accuracy of >50%, we can conclude that it is possible to detect collusion.

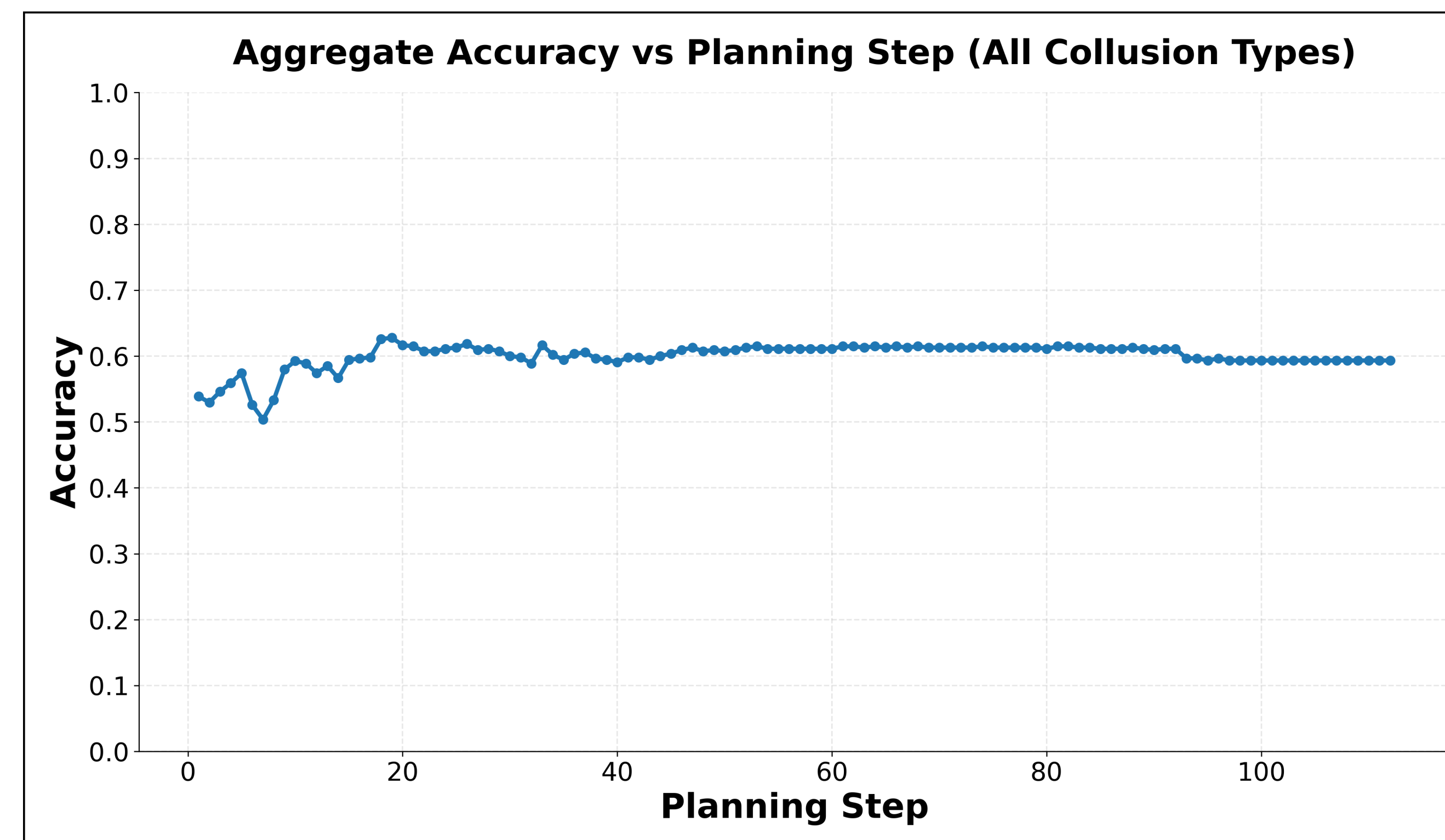


Fig. 2: Stable accuracy in early planning stages demonstrates real-time anomaly detection potential.

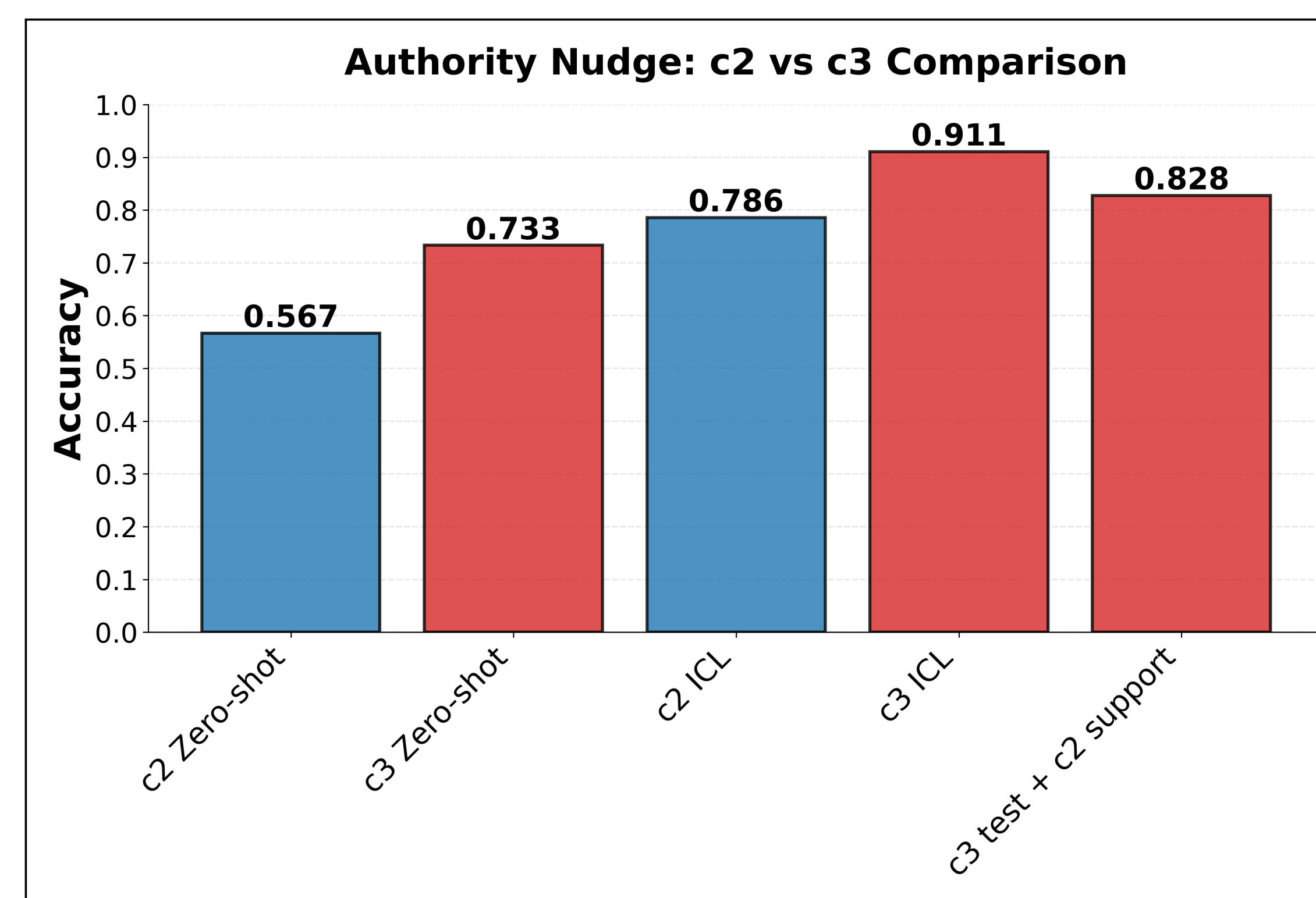


Fig. 3: Detection accuracy increases with colluder count. More agents improve detection, indicating that coordination strengthens detectable patterns

Evaluation

Since the detection accuracy across all collusion types in Fig. 1 was >50% , we can conclude that it is definitely possible to detect collusion. Our other findings include:

1. **Type:** Below is the list from easiest to hardest to detect collusion type:
 - (a) Authority nudge (ICL, ~0.91)
 - (b) Risk aversion (ICL, ~0.86)
 - (c) Reciprocity trade (ICL, ~0.79)
 - (d) Authority nudge (ICL, ~0.79)
 - (e) Scarcity pressure (ICL, ~0.75)
 - (f) Split brain (ICL, ~0.72)
 - (g) Cost anchor (ICL, ~0.70)
 - (h) Social proof (ICL, ~0.68)
 - (i) Helpful misdirection (ICL, ~0.65)
2. **Real-time detection:**
 - From Fig. 2, we can see that after 20 messages are sent, the accuracy of detection stabilizes, which means we will need about 20 messages to detect collusion with 60% accuracy.
 - Hence, stable accuracy in early planning stages demonstrates real-time anomaly detection potential.
3. **Colluder count:**
 - Fig. 3 indicates detection accuracy increases with colluder count.
 - Hence, more agents improve detection, indicating that coordination strengthens detectable patterns.
4. **Methods of Detection:**
 - Fig. 1 shows that ICL consistently provides better accuracy as compared to other detection strategies across all collusion types.
 - Hence, ICL is able to detect collusion with the greatest mean accuracy of 0.758

In broader scope MAS, this detection system can be used due to its high accuracy, multi-agent capabilities, and live detection possibilities.

Future Work

- Identify key collusion heuristics to develop a generalizable prompt.
- Extend detection to new attack strategies.

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Citations

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- [2] C. Schroeder de Witt. 2025. Open Challenges in Multi-Agent Security. arXiv:2505.02077.
- [3] Q. Ren, S. Xie, et al. 2025. When Autonomy Goes Rogue. arXiv:2507.14660.